



Artificial Intelligence

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UNIT-7

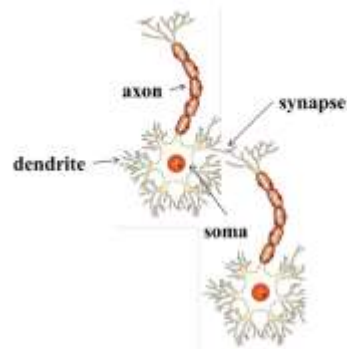
Neural Networks & Expert Systems



Neural Networks

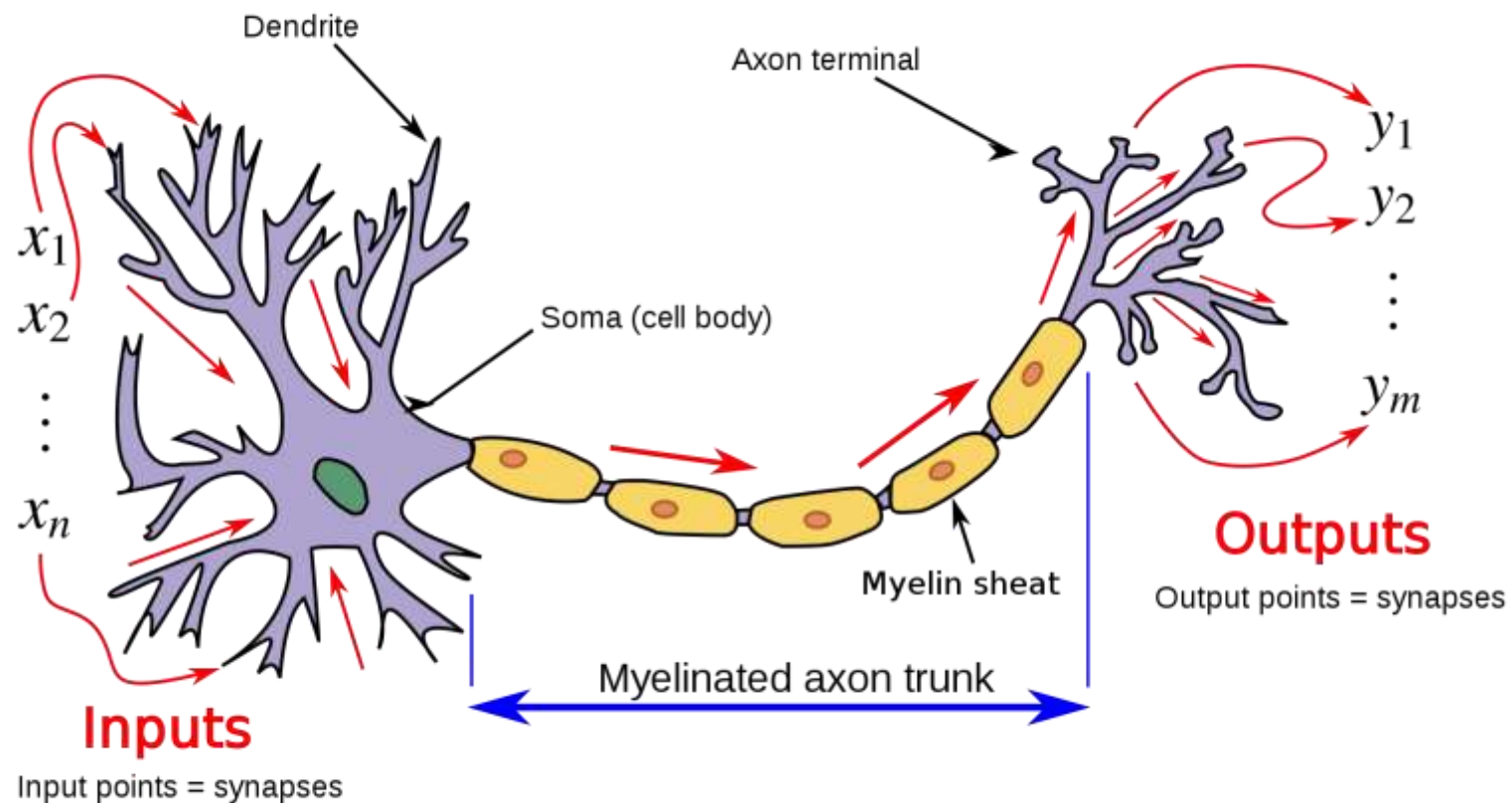
What is Neuron?

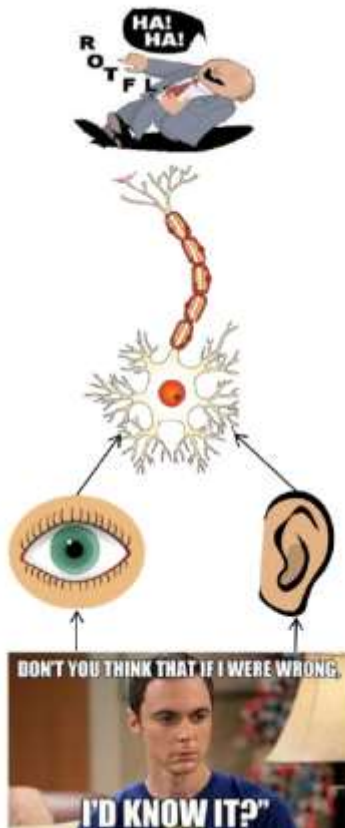
- Neurons are nerve cells that send messages all over your body to allow you to do everything from breathing to talking, eating, walking, and thinking.



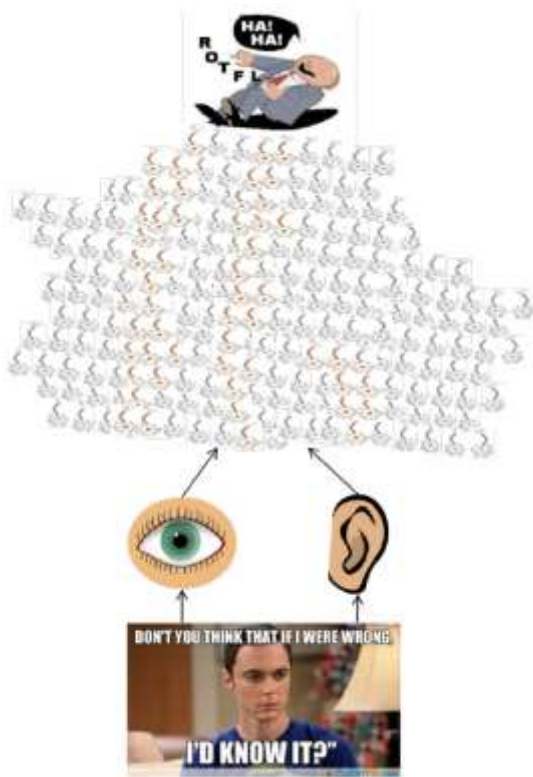
Biological Neurons*

- **dendrite:** receives signals from other neurons
- **synapse:** point of connection to other neurons
- **soma:** processes the information
- **axon:** transmits the output of this neuron

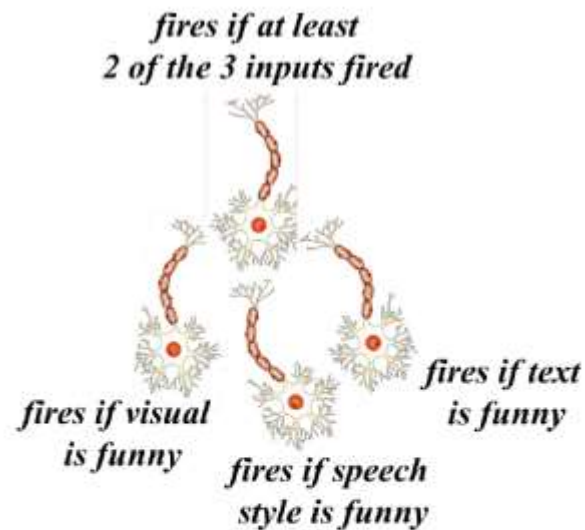




- Let us see a very cartoonish illustration of how a neuron works
- Our sense organs interact with the outside world
- They relay information to the neurons
- The neurons (may) get activated and produces a response (laughter in this case)

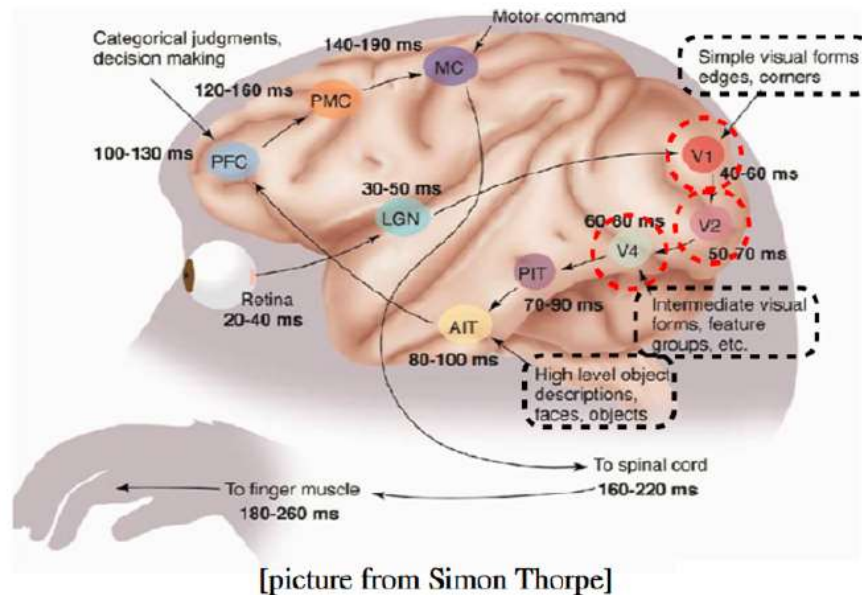


- Of course, in reality, it is not just a single neuron which does all this
- There is a massively parallel interconnected network of neurons
- The sense organs relay information to the lowest layer of neurons
- Some of these neurons may fire (in red) in response to this information and in turn relay information to other neurons they are connected to
- These neurons may also fire (again, in red) and the process continues eventually resulting in a response (laughter in this case)
- An average human brain has around 10^{11} (100 billion) neurons!

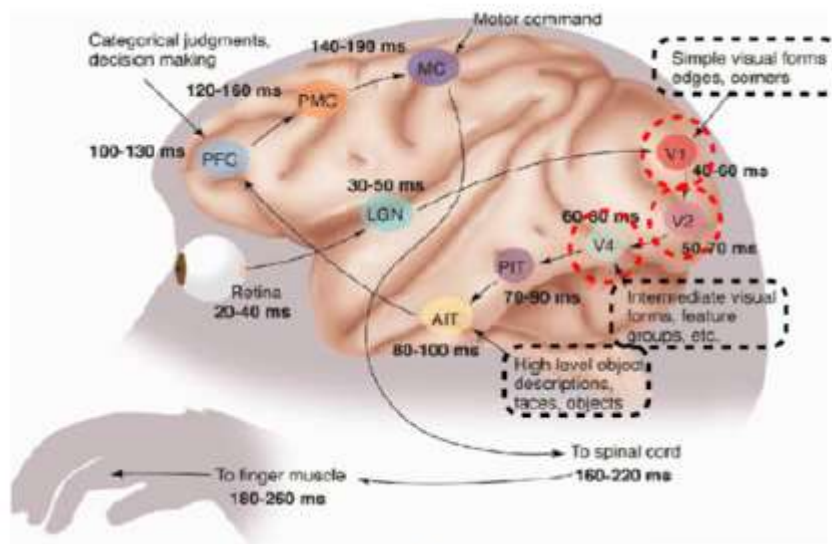


A simplified illustration

- This massively parallel network also ensures that there is division of work
- Each neuron may perform a certain role or respond to a certain stimulus



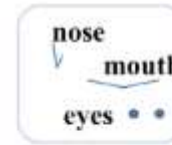
- The neurons in the brain are arranged in a hierarchy
- We illustrate this with the help of visual cortex (part of the brain) which deals with processing visual information
- Starting from the retina, the information is relayed to several layers (follow the arrows)
- We observe that the layers V1, V2 to AIT form a hierarchy (from identifying simple visual forms to high level objects)



[picture from Simon Thorpe]



Layer 1: detect edges & corners



Layer 2: form feature groups

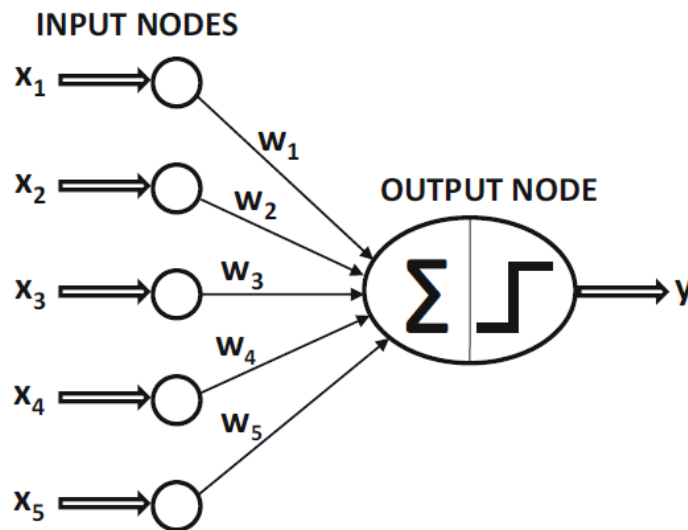


Layer 3: detect high level objects, faces, etc.

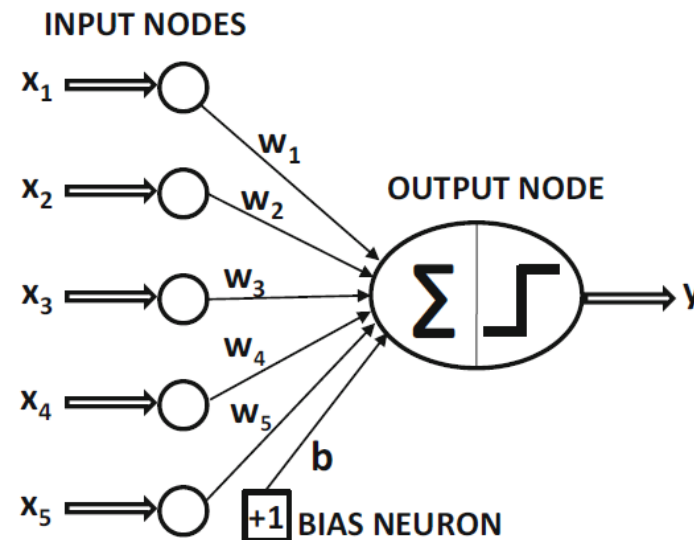
Sample illustration of hierarchical processing*

Artificial Neurons

- Artificial neuron is the most fundamental unit of Deep Neural Network.



(a) Perceptron without bias



(b) Perceptron with bias

Dendrites from Biological Neural Network represent inputs in Artificial Neural Networks, cell nucleus represents Nodes, synapse represents Weights, and Axon represents Output.

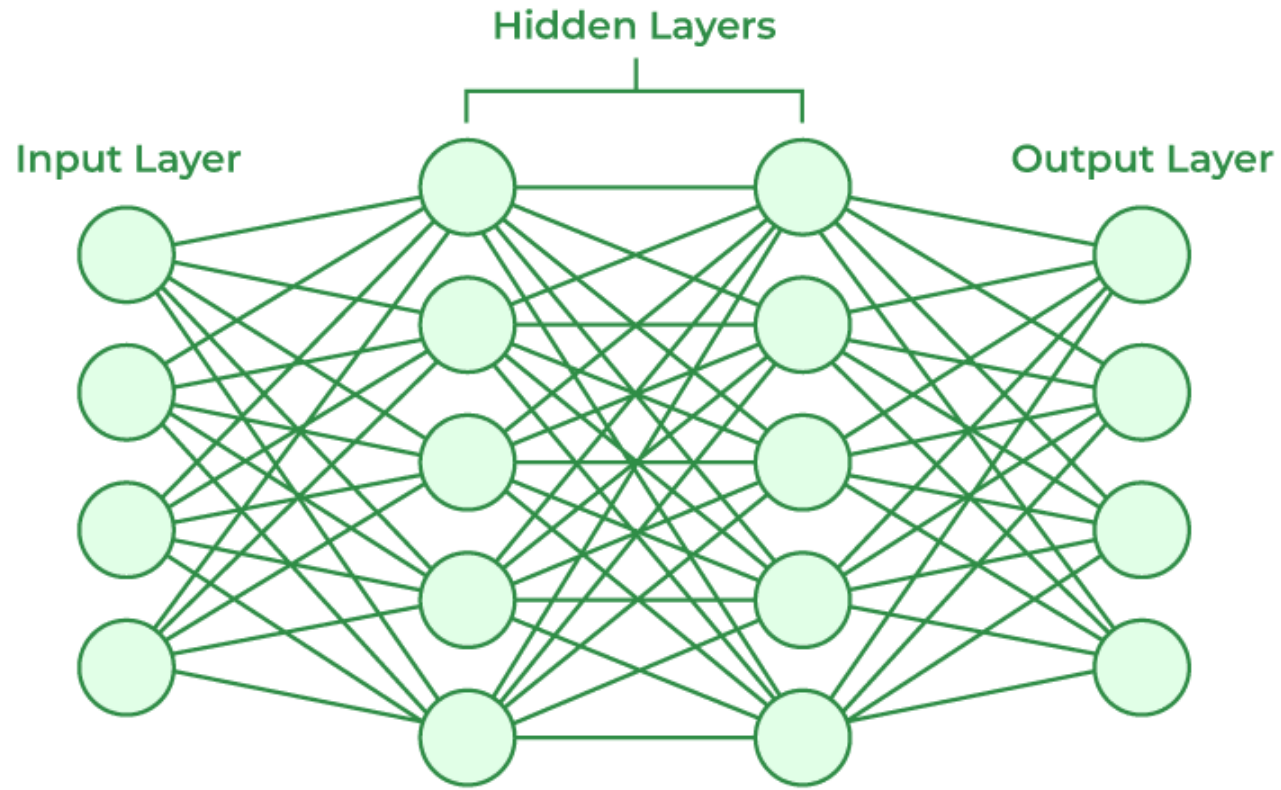
Artificial Neural Network

An **Artificial Neural Network** in the field of **Artificial intelligence** where it attempts to mimic the network of neurons that makes up a human brain so that computers will have an option to understand things and make decisions in a human-like manner. The artificial neural network is designed by programming computers to behave simply like interconnected brain cells.

Architecture of ANN

- To understand the concept of the architecture of an artificial neural network, we have to understand what a neural network consists of.
- In order to define a neural network that consists of a large number of artificial neurons, which are termed units arranged in a sequence of layers.
- Lets us look at various types of layers available in an artificial neural network.
- Artificial Neural Network primarily consists of three layers: Input Layer, Hidden Layer and Output Layer

Architecture of ANN



Input Layer:

As the name suggests, it accepts inputs in several different formats provided by the programmer.

Hidden Layer:

The hidden layer presents in-between input and output layers. It performs all the calculations to find hidden features and patterns.

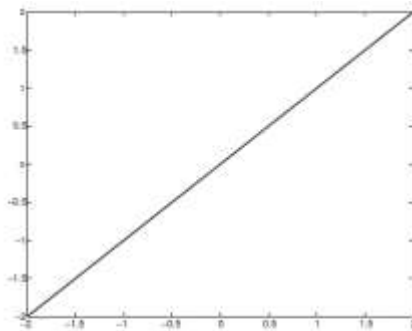
Output Layer:

The input goes through a series of transformations using the hidden layer, which finally results in output that is conveyed using this layer.

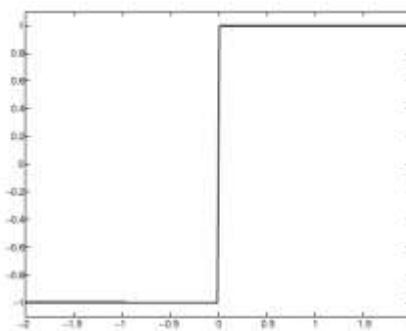
The artificial neural network takes input and computes the weighted sum of the inputs and includes a bias. This computation is represented in the form of a transfer function.

It determines weighted total is passed as an input to an activation function to produce the output. Activation functions choose whether a node should fire or not. Only those who are fired make it to the output layer. There are distinctive activation functions available that can be applied upon the sort of task we are performing.

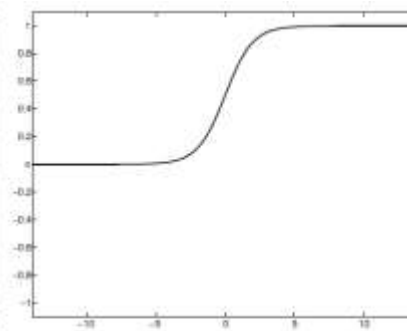
Activation Functions



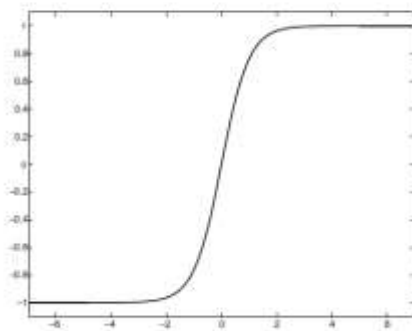
(a) Identity



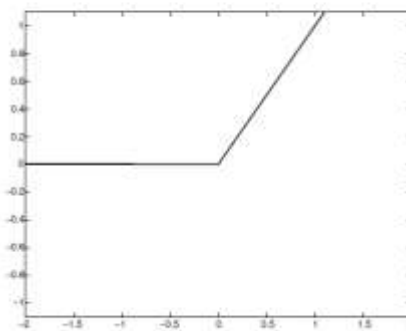
(b) Sign



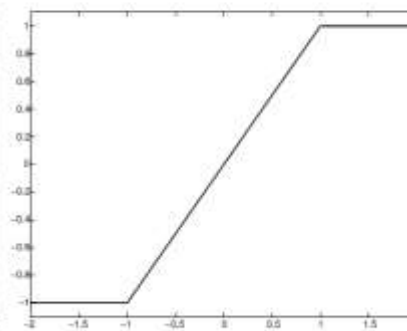
(c) Sigmoid



(d) Tanh



(e) ReLU



(f) Hard Tanh

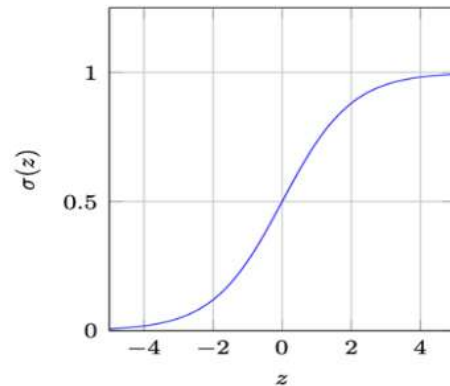
Choice of Activation Function

- ❑ Choosing the right activation function in deep learning is essential for the network to learn effectively.
- ❑ Choosing the right activation function often involves experimentation and depends on factors like the architecture of your network, the nature of your data, and the specific task you're trying to solve. As a general rule of thumb, start with ReLU and its variants (Leaky ReLU, PReLU) and experiment with other activation functions if needed based on performance on a validation set.

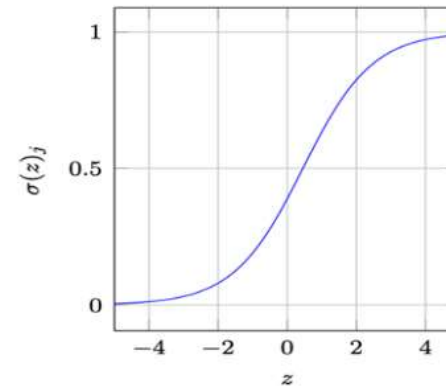
1. **Sigmoid:** Historically used in the output layer for binary classification problems where the output is between 0 and 1. However, it has vanishing gradients problem which can slow down learning in deeper networks.
2. **Tanh (Hyperbolic Tangent):** Similar to sigmoid but ranges from -1 to 1. It is also prone to the vanishing gradient problem, but it tends to work better than sigmoid in hidden layers because its output is zero-centered.
3. **ReLU (Rectified Linear Unit):** It has become the most commonly used activation function in deep learning due to its simplicity and effectiveness. It replaces all negative values with zero, allowing the network to learn faster. However, it can suffer from the "dying ReLU" problem where neurons get stuck at zero and stop learning.
4. **Leaky ReLU:** A variation of ReLU that allows a small gradient when the unit is not active (i.e., when $x < 0$). This addresses the dying ReLU problem by allowing some gradient flow for negative inputs.
5. **Parametric ReLU (PReLU):** Similar to Leaky ReLU but allows the slope of the

6. ELU (Exponential Linear Unit): Similar to ReLU but with a smoothed version for negative inputs, which helps with the vanishing gradient problem. However, it's computationally more expensive than ReLU.

7. Softmax
It normalizes
classes



(a) Sigmoid activation function.



(b) Softmax activation function.

Figure 1: Sigmoid and Softmax activation functions

problems.
le

Leaky rectified linear unit (Leaky ReLU) ^[17]		$f(x) = \begin{cases} 0.01x & \text{for } x < 0 \\ x & \text{for } x \geq 0 \end{cases}$
Parameteric rectified linear unit (PReLU) ^[18]		$f(\alpha, x) = \begin{cases} \alpha x & \text{for } x < 0 \\ x & \text{for } x \geq 0 \end{cases}$
Randomized leaky rectified linear unit (RReLU) ^[19]		$f(\alpha, x) = \begin{cases} \alpha x & \text{for } x < 0 \\ x & \text{for } x \geq 0 \end{cases}$ ^[3]
Exponential linear unit (ELU) ^[20]		$f(\alpha, x) = \begin{cases} \alpha(e^x - 1) & \text{for } x \leq 0 \\ x & \text{for } x > 0 \end{cases}$
Scaled exponential linear unit (SELU) ^[21]		$f(\alpha, x) = \lambda \begin{cases} \alpha(e^x - 1) & \text{for } x < 0 \\ x & \text{for } x \geq 0 \end{cases}$ with $\lambda = 1.0507$ and $\alpha = 1.67326$

How do Artificial Neural Networks learn?

- Artificial neural networks are trained using a training set. For example, suppose you want to teach an ANN to recognize a cat. Then it is shown thousands of different images of cats so that the network can learn to identify a cat.
- Once the neural network has been trained enough using images of cats, then you need to check if it can identify cat images correctly. This is done by making the ANN classify the images it is provided by deciding whether they are cat images or not.
- The output obtained by the ANN is corroborated by a human-provided description of whether the image is a cat image or not.
- If the ANN identifies incorrectly then [back-propagation](#) is used to adjust whatever it has learned during training.

How do Artificial Neural Networks learn?

- [Backpropagation](#) is done by fine-tuning the weights of the connections in ANN units based on the error rate obtained. This process continues until the artificial neural network can correctly recognize a cat in an image with minimal possible error rates.

Types of ANN

1. Feed forward Neural Network (FFNN)
2. Convolution Neural Network (CNN)
3. Recurrent Neural Network (RNN)

Feed forward Neural Network (FFNN)

- The feedforward neural network is one of the most basic artificial neural networks. In this ANN, the data or the input provided travels in a single direction.
- It enters into the ANN through the input layer and exits through the output layer while hidden layers may or may not exist.
- So the feedforward neural network has a front-propagated wave only.

Convolution Neural Network (CNN)

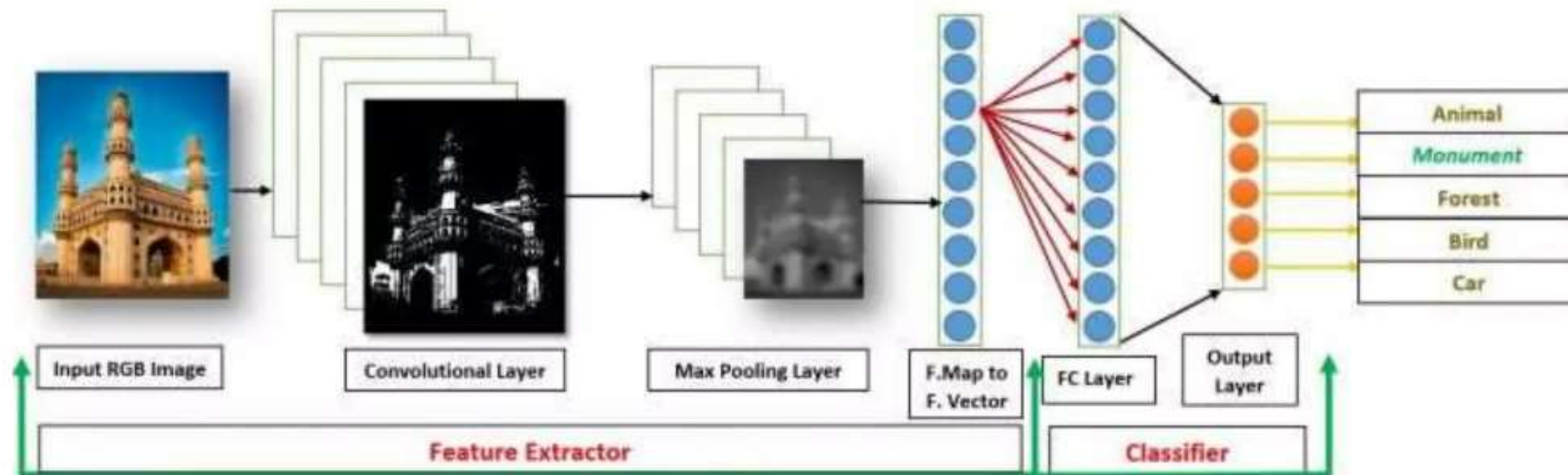
- A Convolutional neural network has some similarities to the feed-forward neural network, where the connections between units have weights that determine the influence of one unit on another unit.
- But a CNN has one or more than one convolutional layer that uses a convolution operation on the input and then passes the result obtained in the form of output to the next layer.
- CNN has applications in speech and image processing which is particularly useful in computer vision.

Basic Components of CNN

Convolution Layer

Pooling Layer

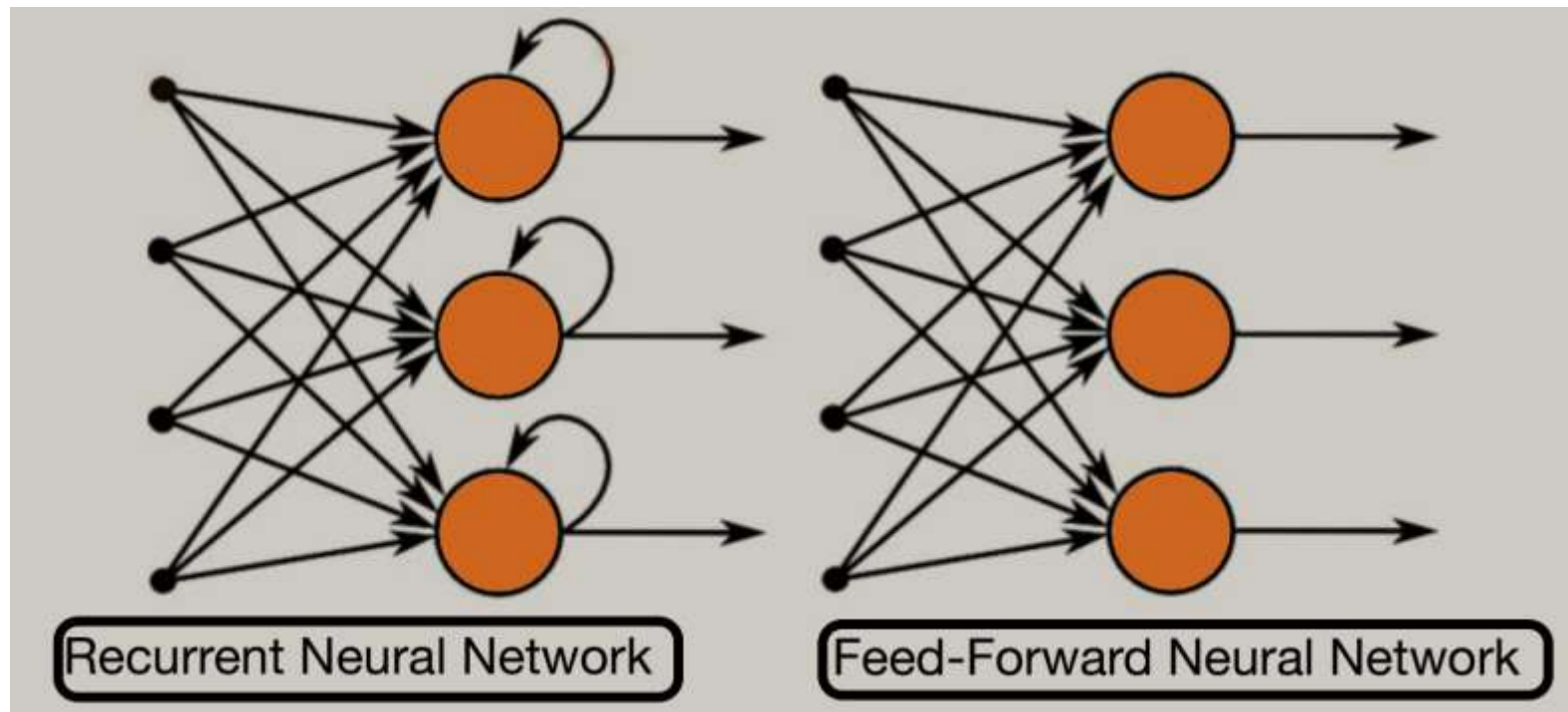
Output layer or fully connected layer



Recurrent Neural Network (RNN)

- A recurrent neural network (RNN) is a type of artificial neural network designed to recognize patterns in sequences of data, such as time series, speech, text, financial data, etc.
- The principle of Recurrent Neural Network is to feedback the output of a layer back to the input again. This principle helps to predict the outcome of the layer.
- In the Computation process, Each neuron will act as a memory cell. The neuron will retain some information as it goes to the next time step.
- It is called a recurrent neural network process. The data to be used later will be remembered and work for the next step will go on in the process.

RNN vs Feed Forward Neural Network



A recurrent neural network appears very just like feedforward neural networks, except it also has connections pointing backwards.

Applications of Neural Networks

They can perform tasks that are easy for a human but difficult for a machine –

- **Aerospace** – Autopilot aircrafts, aircraft fault detection.
- **Automotive** – Automobile guidance systems.
- **Military** – Weapon orientation and steering, target tracking, object discrimination, facial recognition, signal/image identification.
- **Electronics** – Code sequence prediction, IC chip layout, chip failure analysis, machine vision, voice synthesis.
- **Financial** – Real estate appraisal, loan advisor, mortgage screening, corporate bond rating, portfolio trading program, corporate financial analysis, currency value prediction, document readers, credit application evaluators.

Industrial – Manufacturing process control, product design and analysis, quality inspection systems, welding quality analysis, paper quality prediction, chemical product design analysis, dynamic modeling of chemical process systems, machine maintenance analysis, project bidding, planning, and management.

Medical – Cancer cell analysis, EEG and ECG analysis, prosthetic design, transplant time optimizer.

Speech – Speech recognition, speech classification, text to speech conversion.

Telecommunications – Image and data compression, automated information services, real-time spoken language translation.

Transportation – Truck Brake system diagnosis, vehicle scheduling, routing systems.

Software – Pattern Recognition in facial recognition, optical character recognition, etc.

- **Time Series Prediction** – ANNs are used to make predictions on stocks and natural calamities.
- **Signal Processing** – Neural networks can be trained to process an audio signal and filter it appropriately in the hearing aids.
- **Control** – ANNs are often used to make steering decisions of physical vehicles.
- **Anomaly Detection** – As ANNs are expert at recognizing patterns, they can also be trained to generate an output when something unusual occurs that misfits the pattern.

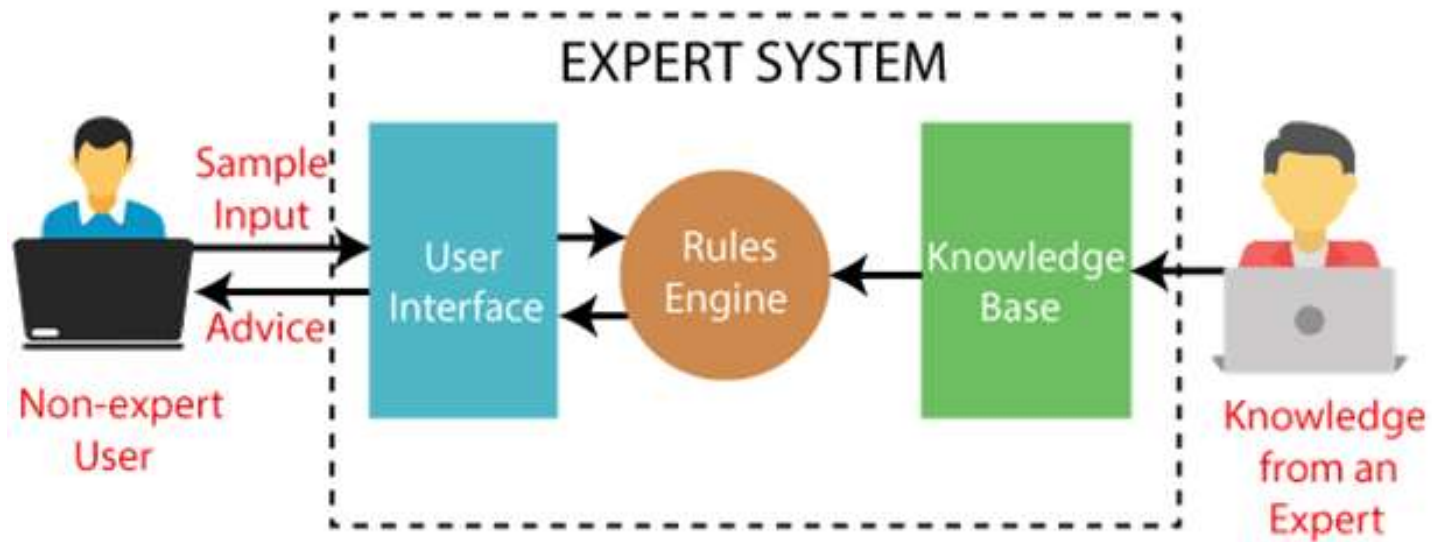
Expert Systems

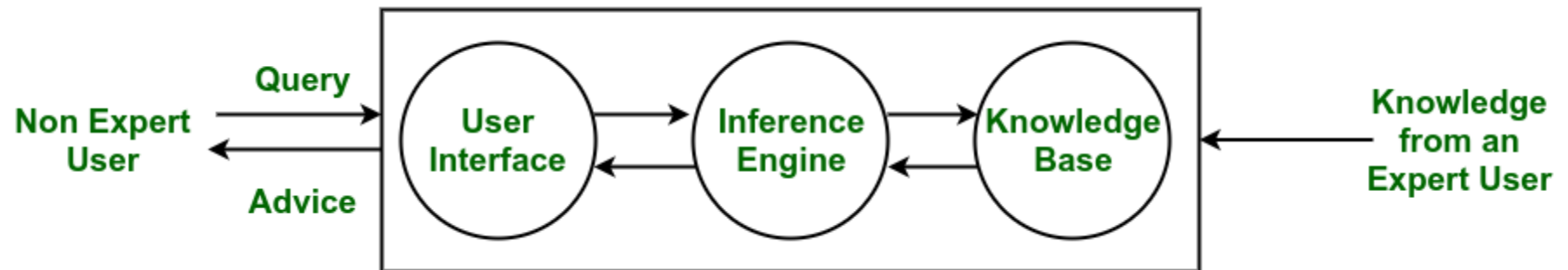
Expert Systems (ES)

- An expert system is a computer program or AI software that is designed to solve complex problems and to provide decision-making ability like a human expert.
- It performs this by extracting knowledge from its knowledge base using the reasoning and inference rules according to the user queries.
- The data in the knowledge base is added by humans that are expert in a particular domain and this software is used by a non-expert user to acquire some information.
- It is called so because it contains the expert knowledge of a specific domain and can solve any complex problem of that particular domain.
- These systems are designed for specific domain such as medical diagnosis, accounting, coding, games etc.

- The performance of an expert system is based on the expert's knowledge stored in its knowledge base (KB).
- The more knowledge stored in the KB, the more that system improves its performance. One of the common examples of an ES is a suggestion of spelling errors while typing in the Google search box.

Note: It is important to remember that an expert system is not used to replace the human experts; instead, it is used to assist the human in making a complex decision. These systems do not have human capabilities of thinking and work on the basis of the knowledge base of the particular domain.





Components of an expert system:

Knowledge base: The knowledge base represents facts and rules. It consists of knowledge in a particular domain as well as rules to solve a problem, procedures and intrinsic data relevant to the domain.

Inference engine: The function of the inference engine is to fetch the relevant knowledge from the knowledge base, interpret it and to find a solution relevant to the user's problem. The inference engine acquires the rules from its knowledge base and applies them to the known facts to infer new facts. Inference engines can also include an explanation and debugging abilities.

Knowledge acquisition and learning module: The function of this component is to allow the expert system to acquire more and more knowledge from various sources and store it in the knowledge base.



Components of an expert system:

User interface: This module makes it possible for a non-expert user to interact with the expert system and find a solution to the problem.

Explanation module: This module helps the expert system to give the user an explanation about how the expert system reached a particular conclusion.



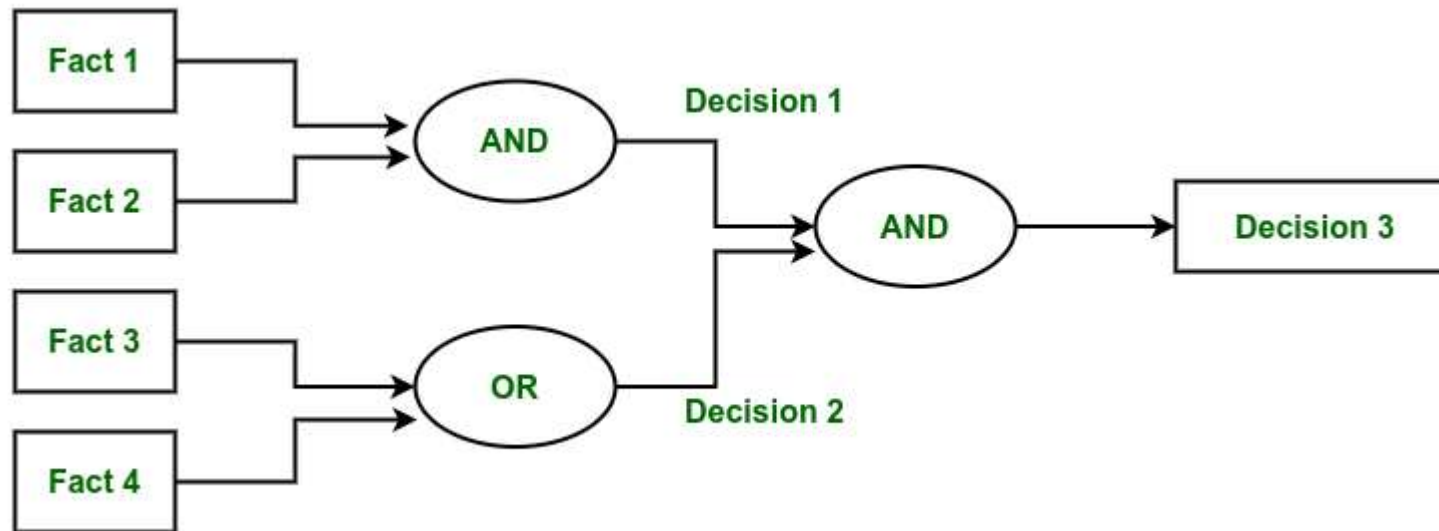
The Inference Engine generally uses two strategies for acquiring knowledge from the Knowledge Base, namely –

- Forward Chaining
- Backward Chaining

Forward Chaining:

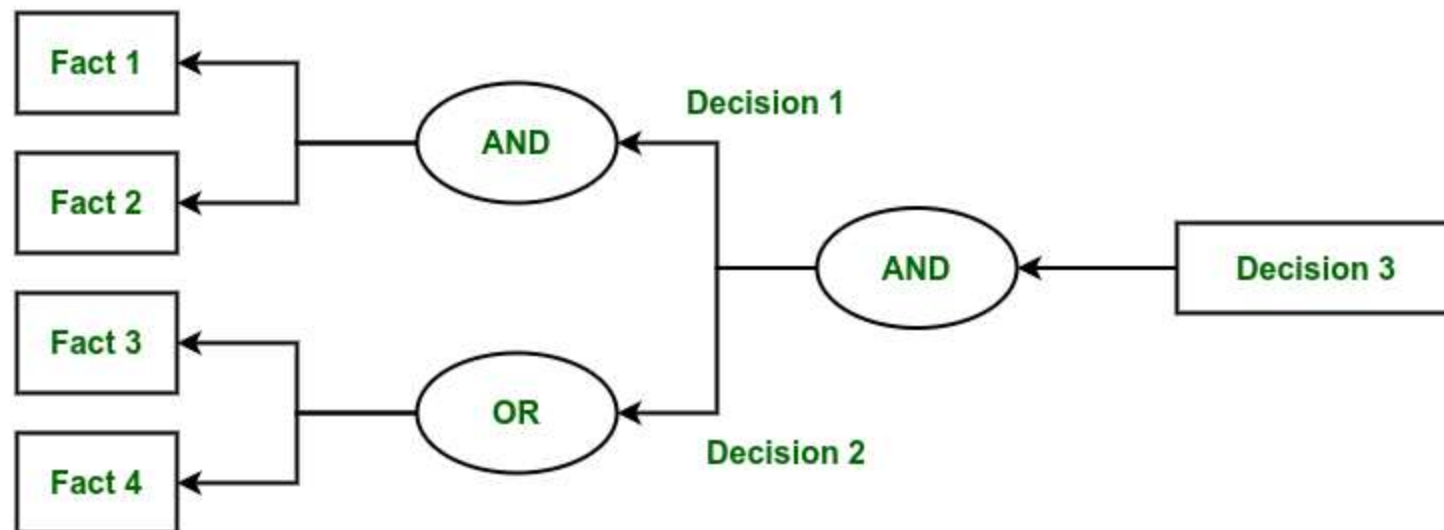
Forward Chaining is a strategic process used by the Expert System to answer the questions – What will happen next. This strategy is mostly used for managing tasks like creating a conclusion, result or effect. Example – prediction or share market movement status.

Forward Chaining



Backward Chaining

Backward Chaining is a strategy used by the Expert System to answer the questions – Why this has happened. This strategy is mostly used to find out the root cause or reason behind it, considering what has already happened. Example – diagnosis of stomach pain, blood cancer or dengue, etc.



Characteristics of Expert System in AI

- Human experts are perishable but an expert system is permanent.
- It helps to distribute the expertise of a human.
- One expert system may contain knowledge from more than one human experts thus making the solutions more efficient.
- It decreases the cost of consulting an expert for various domains such as medical diagnosis.
- They use a knowledge base and inference engine.



Characteristics of Expert System in AI

- Expert systems can solve complex problems by deducing new facts through existing facts of knowledge, represented mostly as if-then rules rather than through conventional procedural code.
- Expert systems were among the first truly successful forms of artificial intelligence (AI) software.

Capabilities of Expert System in AI

- Advising
- Instructing and assisting a human in decision making
- Demonstrating
- Deriving a solution
- Diagnosing
- Explaining
- Interpreting input
- Predicting results
- Justifying the conclusion



Advantages of Expert Systems (ES)

- Low accessibility cost.
- Fast response.
- Not affected by emotions, unlike humans.
- Low error rate.
- Capable of explaining how they reached a solution.



Disadvantages of Expert Systems(ES)

- Expert system have no emotions.
- Common sense is the main issue of the expert system.
- It is developed for a specific domain.
- It needs to be updated manually. It does not learn itself.
- Not capable to explain the logic behind the decision



Examples of Expert Systems(ES)

MYCIN: It was one of the earliest backward chaining expert systems that was designed to find the bacteria causing infections like bacteraemia and meningitis. It was also used for the recommendation of antibiotics and the diagnosis of blood clotting diseases.

DENDRAL: It was an artificial intelligence project that was made as a chemical analysis expert system. It was used in organic chemistry to detect unknown organic molecules with the help of their mass spectra and knowledge base of chemistry.

CaDeT: The CaDet expert system is a diagnostic support system that can detect cancer at early stages.

PXDES: It is an expert system that is used to determine the type and level of lung cancer. To determine the disease, it takes a picture from the upper body, which looks like the shadow. This shadow identifies the type and degree of harm.

- The strength of an ES derives from its ***knowledge base*** - an organized collection of facts and heuristics about the system's domain.
- An ES is built in a process known as ***knowledge engineering***, during which knowledge about the domain is acquired from human experts and other sources by knowledge engineers.
- ***Knowledge representation*** is the method used to organize the knowledge in the knowledge base.
- Knowledge representation is not just storing data into some database, but it also enables an intelligent machine to learn from that knowledge and experiences so that it can behave intelligently like a human.

- **Knowledge Representation** is a key aspect of expert systems and involves formalizing knowledge in a way that allows the system to reason and make decisions.
- Several techniques are used for **knowledge representation in AI**. This includes logical representation, semantic networks, frame representation, and production rules.

1. Rule-Based Systems (Production Rules)

- **IF-THEN Rules:** The most common form of knowledge representation. Each rule consists of an antecedent (IF part) and a consequent (THEN part). For example, "IF the engine won't start AND the battery is dead THEN the problem is the battery".
- **Forward Chaining:** Starts with known facts and applies rules to infer new facts.
- **Backward Chaining:** Starts with a goal and works backward to determine the facts that support the goal.

2. Logical Representation involves using formal languages such as propositional logic, first-order logic, and predicate calculus to represent facts and relationships. This allows the system to apply logical reasoning to arrive at conclusions. For example, an expert system in medical diagnosis might use logical representation to infer a specific disease based on symptoms and medical history.

3. Semantic Networks provide a graphical representation of concepts and their relationships. Nodes represent concepts, while links depict relationships between concepts. This technique is useful for representing hierarchical structures and complex relationships.

For instance, an expert system for natural language processing might use a semantic network to represent the relationships between words in a sentence.

4. Frame Representation

Frame Representation involves organizing knowledge into frames that represent objects, concepts, or situations with attributes and slots. Frames capture structured information about entities in a domain and allow reasoning based on these attributes and slots. An example of frame-based representation is the use of frames to represent different car models, where each frame contains attributes like colour, engine type, and price range.

Another Example: A "restaurant" frame might include slots for "menu," "location," "cuisine type," etc.

- **Advantages:** Excellent for representing structured knowledge and default values
- **Disadvantages:** Can be inflexible and difficult to adapt to new information.

Background: MYCIN was an early expert system developed in the 1970s at Stanford University. It was designed to assist physicians in diagnosing bacterial infections and recommending appropriate antibiotics.

Knowledge Representation: MYCIN used a rule-based knowledge representation, with its knowledge encoded in the form of "if-then" production rules. The system contained over 450 rules, covering various aspects of infectious diseases, symptoms, lab results, and drug interactions.

Implementation:

- **Inference Engine:** MYCIN's inference engine applied backward chaining, starting with a goal (e.g., identifying the causative agent of an infection) and working backward through the rules to gather necessary data.
- **User Interface:** The system interacted with users (physicians) through a question-and-answer format, gathering information about the patient's symptoms, medical history, and test results.

Results:

- **Accuracy:** Studies showed that MYCIN's diagnostic accuracy was comparable to that of expert physicians. In some cases, MYCIN provided better recommendations by considering more factors systematically.
- **Impact:** MYCIN demonstrated the potential of expert systems to assist in complex decision-making processes, laying the groundwork for future medical AI applications.

Background: XCON (eXpert CONfigurer) was developed by Digital Equipment Corporation (DEC) in the 1980s to assist in configuring VAX computer systems. The complexity of custom configurations required expert knowledge to ensure all components were compatible and optimized.

Knowledge Representation: XCON utilized a rule-based system with hundreds of rules capturing the expertise of DEC engineers. The rules covered hardware compatibility, spatial constraints within computer cabinets, and optimization of system performance.

Implementation:

- Inference Engine:** XCON employed forward chaining to apply rules iteratively, starting with the basic components specified by the customer and adding compatible modules until a complete configuration was achieved.
- User Interface:** Sales and support staff interacted with XCON through a graphical interface that facilitated the input of customer requirements and the visualization of configuration options.

Results:

- Efficiency:** XCON significantly reduced the time required to generate accurate and optimized system configurations, from days or weeks to a matter of minutes.
- Accuracy:** The system improved the accuracy of configurations, reducing errors and the need for costly post-sale adjustments.
- Economic Impact:** XCON saved DEC millions of dollars by streamlining the configuration process and enhancing customer satisfaction with tailored solutions.